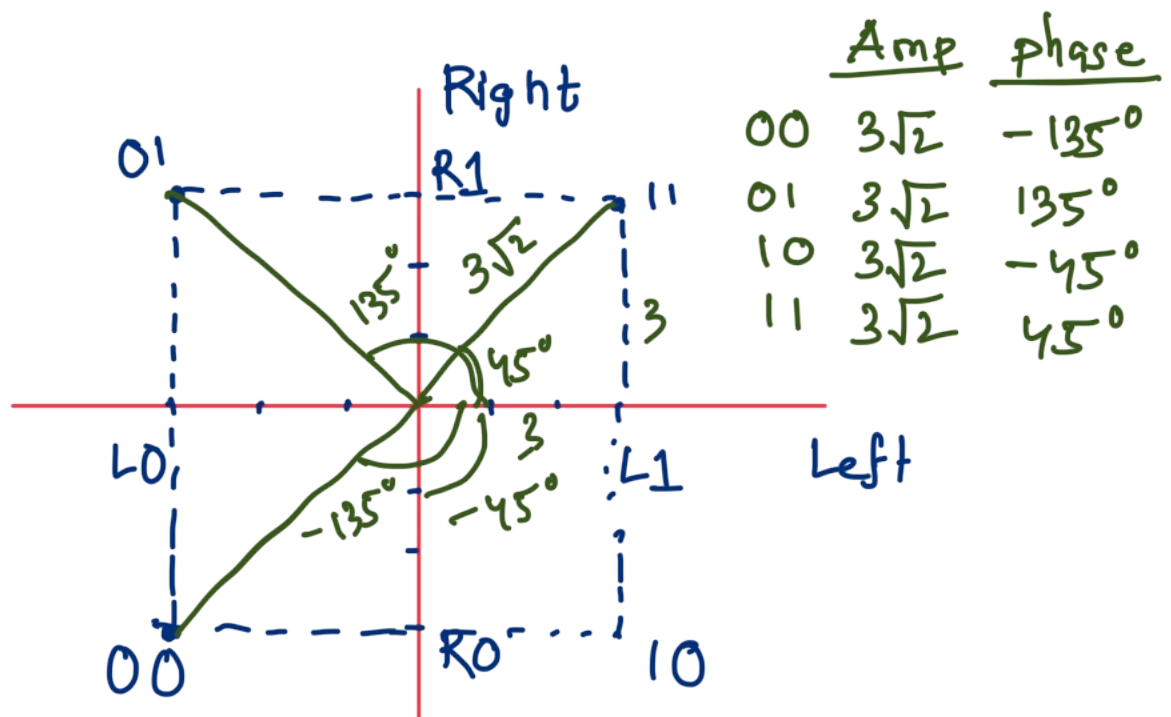
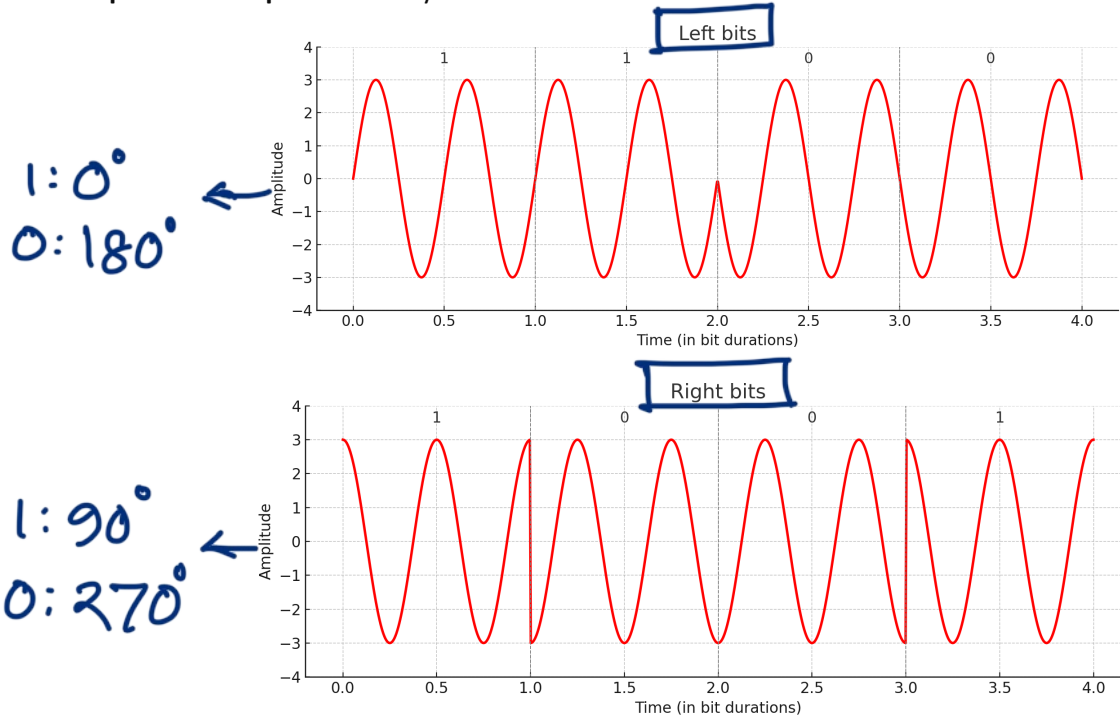
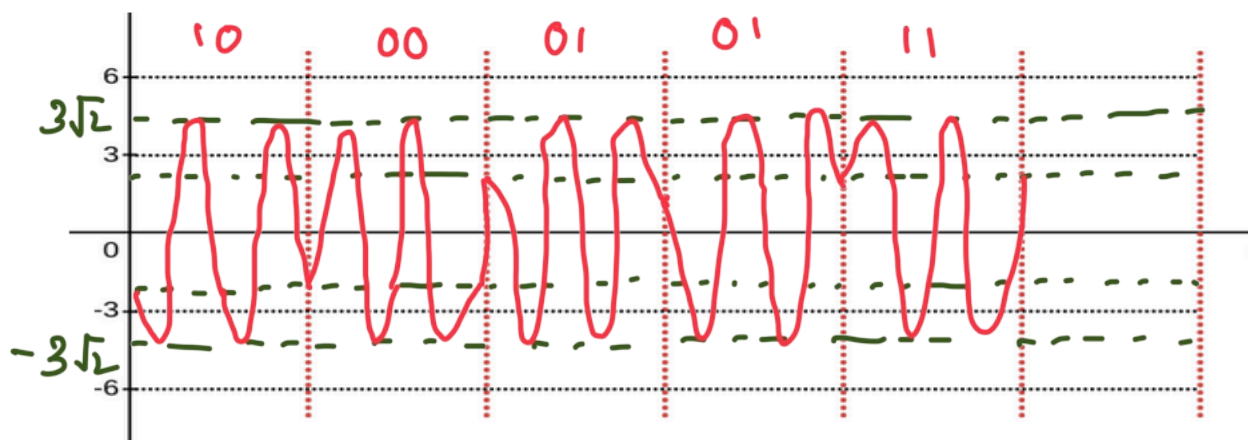


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1. For the QPSK-modulation scheme given below, draw the **constellation diagram** and show the resultant **amplitudes** and **phases** clearly in a table.

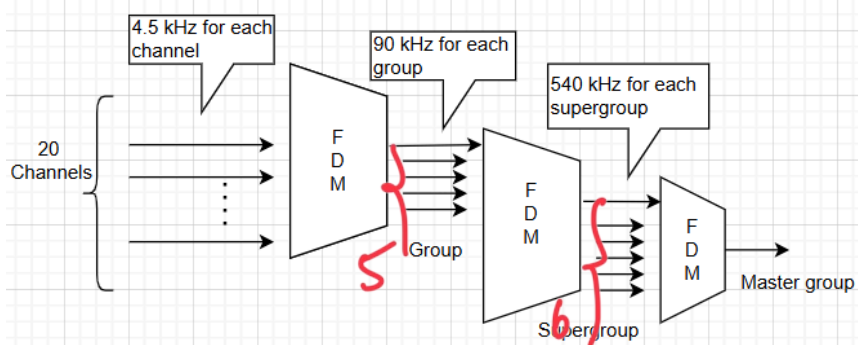


2. Using the QPSK scheme of question 2, draw the modulated signal for the bit stream 1000101111.



3. The following figure shows an analog hierarchical system used by a telephone company for multiplexing. The bandwidth of each link is shown in the box above the links.

Analog Hierarchy:



- Calculate the **total number of channels** in the **master group**. [2]
- Calculate the **guard band** used between the **groups** before forming the supergroup. [3]

A. # channels = $20 \times 5 \times 6 = 600$

B. 5 groups $\rightarrow$ 1 supergroup

$\therefore$ Blw in supergroup = $5 \times 90 = 450$ kHz

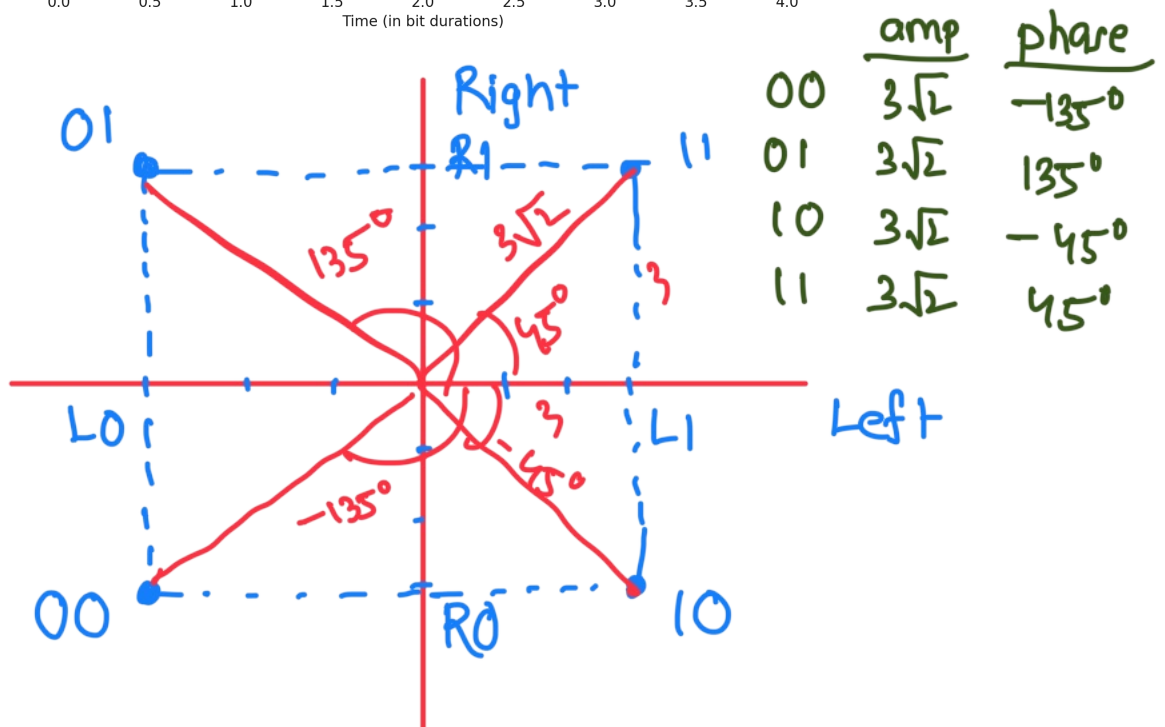
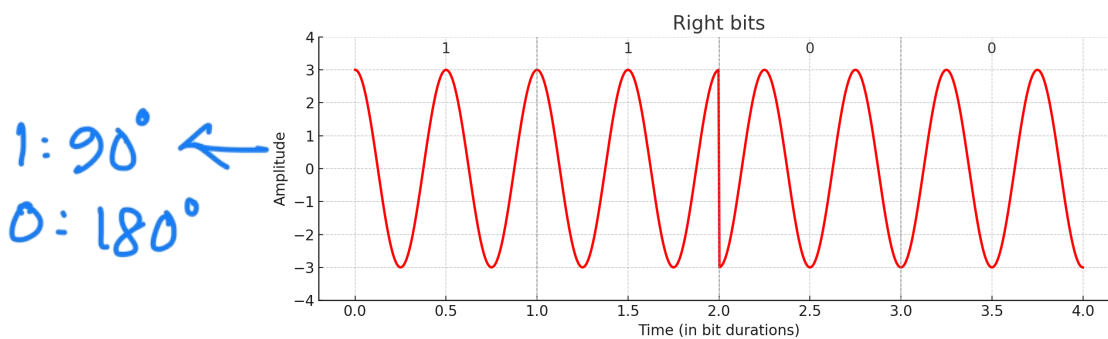
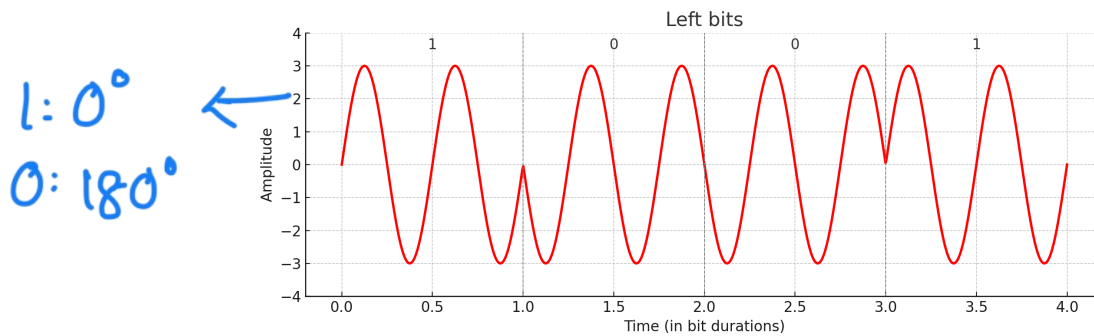
Given blw = 540 kHz

Extra(for guard band) = $540 - 450 = 90$ kHz

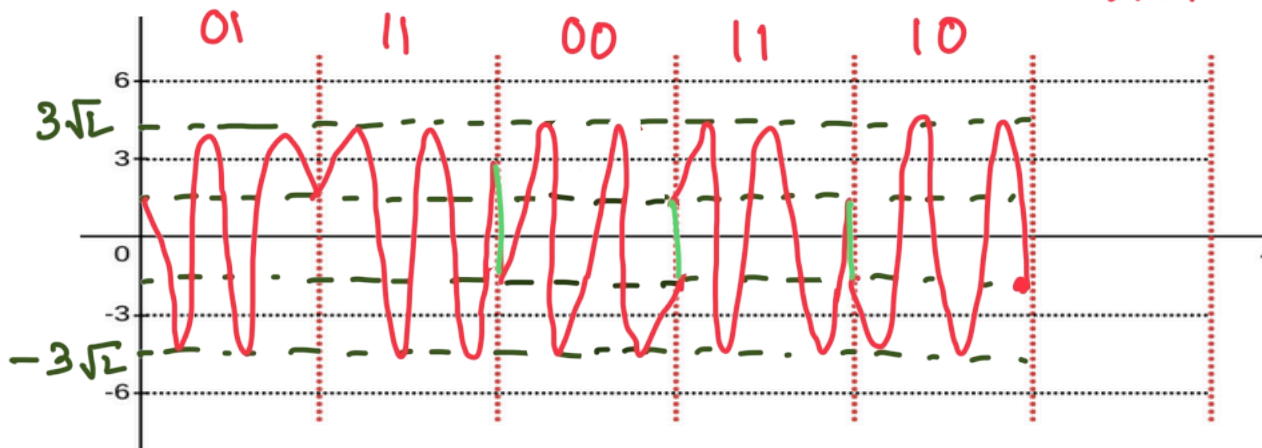
$\therefore$ Each guard band = $\frac{90}{4} = 22.5$ kHz
 $\rightarrow (n-1)$ for n channels

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1. For the QPSK-modulation scheme given below, draw the constellation diagram and show the resultant amplitudes and phases clearly in a table. [5]

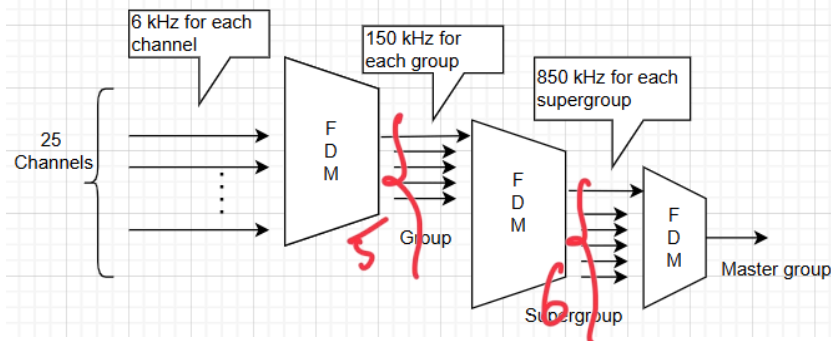


2. Using the QPSK scheme of question 2, draw the modulated signal for the bit stream 0111001110. [5]



3. The following figure shows an analog hierarchical system used by a telephone company for multiplexing. The bandwidth of each link is shown in the box above the links.

Analog Hierarchy:



A. Calculate the **total number of channels** in the **master group**. [2]

B. Calculate the **guard band** used between the **groups** before forming the supergroup. [3]

$$A. \# \text{ channels} = 25 \times 5 \times 6 = 750$$

$$B. \text{ Each guard band} = \frac{850 - 5 \times 150}{4} = 25 \text{ kHz}$$